Risk-Conscious Stock Portfolio Optimization

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TABLE OF CONTENTS

Overview **1**
Methodology & Analysis **1**
Data Used 1
Portfolio Modeling 1
Efficient Frontier 2
Results **2**
Allocation of Optimized Portfolio by Stock 2
Efficient Frontier Portfolio Risk Options 2
Recommendations **4**
Optimized Portfolio Implementation 4
Exploring Efficient Frontier Portfolio Risk Options 4
Conclusion **4**

RISK-CONSCIOUS STOCK PORTFOLIO OPTIMIZATION

Overview

The Commercial Banking Corporation's Portfolio Pricing department seeks our recommendation of a portfolio of five stocks to minimize risk and achieve a daily return of at least 0.18%. Our team was tasked with using the percent daily returns of the closing prices of such stocks, the historical volatility and mean of the percent daily returns, and the development of an efficient frontier to present alternative portfolio strategies for differing levels of risk.

Our team developed a portfolio that had a minimized daily risk of 1.8% while maintaining the budget and a daily return of at least 0.18% by allocating the following percentages of the portfolio to each stock:

- Alphabet Inc. (GOOG): 69.10%
- Apple Inc. (AAPL): 19.66%
- Meta Platforms Inc. (META): 8.77%
- Broadcom Inc. (AVGO): 2.47%
- NVIDIA Corp (NVDA): 0.00%

We recommend that the Bank adopt our optimal portfolio while also exploring our various efficient frontier strategies. Our analysis identified three noteworthy portfolios along the frontier: a low-risk portfolio at 1.7% daily risk with a daily return of 0.14%, a maximum Sharpe ratio portfolio at 2.0% daily risk with a daily return of 0.22%, and a high-risk portfolio at 3.4% daily risk with a daily return of 0.23%.

Methodology & Analysis

This section discusses the dataset provided, how our team prepared it for modeling, and the analyses we performed to identify various stock portfolio options.

Data Used

Our team collected the daily closing prices for the five stocks (NVDA, GOOG, AVGO, AAPL, and META) from Yahoo Finance. We collected these prices for the timeframe from January 2, 2025, through December 31, 2025. Using those closing prices, we calculated the percent daily returns of the closing prices and the summary statistics for individual stocks. These summary statistics included the volatility and mean of the percent daily returns. Our team has also been informed to assume no transaction costs are required to buy or sell stocks, and thus, no additional costs will be incurred.

Portfolio Modeling

Our team created an empty model that we proceeded to add in individual variables that represent each stock. We created our objective function for the model by forming a covariance matrix between the volatility of each stock, indicating the portfolio's risk. We wanted to minimize our objective function so that we ultimately minimize the risk in our portfolio. We added two constraints to our model: the budget constraint and the daily return constraint. The budget constraint sets the summation of the stocks equal to one, as we want to invest all of the Bank's money, while not having to borrow money. The daily return constraint reflected the Bank's desire to minimize risk while maintaining a daily return of at least 0.18%. To do so, our team set the portfolio's daily return mean to be greater than or equal to 0.0018. Once our team had the model set to accommodate the Bank's goals for the portfolio, we performed optimization of the model.

Efficient Frontier

Our team created an efficient frontier graph to visualize alternative portfolio options that may represent varying levels of risk. The bottom half of the efficient frontier graph was disregarded as this region represented risk levels that were not valuable options for the Bank. Our team focused on three key points along the upper frontier: the vertex of the quadratic representing the low-risk portfolio, the maximum Sharpe ratio portfolio representing the most efficient risk-return trade-off at the "knee" of the curve, and the high-risk region in the upper right of the frontier representing a high-return portfolio option.

Results

This section presents findings from optimizing the portfolio model and visualizing various portfolio options of varying risk levels.

Allocation of Optimized Portfolio by Stock

Our team used Gurobi optimization to determine the percentages of the portfolio to allocate to each stock to minimize risk while satisfying the constraints we mentioned previously. **Table 1** below displays the percentages of the portfolio that we determined were best to allocate towards each stock.

Table 1: Allocation of Optimized Portfolio by Individual Stocks

Stock	Percentage of Portfolio
GOOG	69.10%
AAPL	19.66%
META	8.77%
AVGO	2.47%
NVDA	0.00%

The portfolio optimization allocated around 69% to the GOOG stock. This indicated to our team that this stock may be one of the more stable options, minimizing risk for the Bank. The AAPL stock follows GOOG, but not closely, with around 20% of the portfolio allocated to it. On the other hand, the NVDA stock accounted for 0% of the portfolio. This indicated to our team that this stock did not contribute largely to minimizing risk in daily returns, and thus it would not be beneficial for the Bank to allocate any of the portfolio to NVDA.

Efficient Frontier Portfolio Risk Options

Our team created the efficient frontier visualization to identify the three points that represent a low-risk option, a maximum Sharpe ratio option, and a high-risk portfolio option for the Bank. The efficient frontier visualization is illustrated in **Figure 1**.

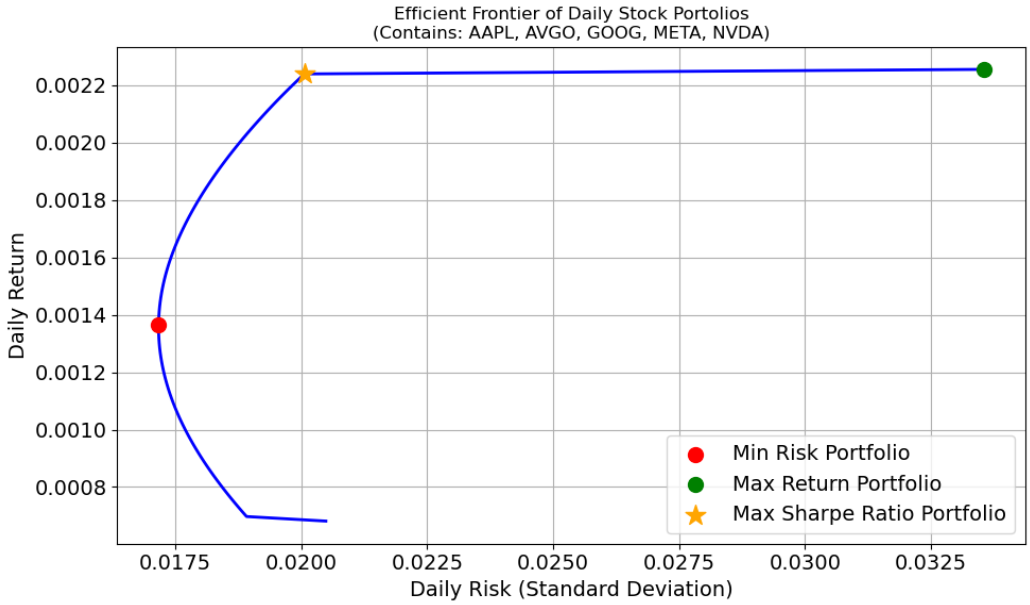

Figure 1: Efficient Frontier of Daily Stock Portfolios

As shown in **Figure 1**, the low-risk option has a risk level of 0.017 (1.7%) and an expected daily return of 0.14%. The maximum Sharpe ratio portfolio, which represents the maximum risk-adjusted return on the efficient frontier, occurs at a risk level of 0.020 (2.0%) and a daily return of 0.22%. Beyond this point, the frontier begins to plateau, and additional risk yields diminishing returns. The high-risk option has a risk level of 0.034 (3.4%) and a daily return of 0.23%.

The three portfolio options we identified have varying portfolio allocations for the five stocks. These portfolio allocations are presented in **Table 2** below.

Table 2: Allocation of Efficient Frontier Portfolios by Individual Stocks

Stock	Percentage of Low-Risk Portfolio	Percentage of High-Risk Portfolio	Percentage of Max Sharpe Ratio Portfolio
AAPL	37.46%	0.00%	0.01%
AVGO	0.00%	100%	7.60%
GOOG	43.40%	0.00%	92.38%
META	19.14%	0.00%	0.00%
NVDA	0.00%	0.00%	0.00%

Our team identified that the high-risk portfolio was 100% allocated towards the stock AVGO, whilst the low-risk portfolio allocates 0.00% of the portfolio towards AVGO. This indicated to our team that AVGO is a high-risk stock that could deliver high returns for the portfolio. Notably, the maximum Sharpe ratio portfolio, the most efficient point on the frontier, allocates 92.38% to GOOG and 7.60% to AVGO, with effectively no allocation to the remaining three stocks. This portfolio also reveals that a small allocation to AVGO can enhance returns without the extreme risk of a 100% AVGO position. The low-risk portfolio, by contrast, allocates about 43% to GOOG and 37% to AAPL, prioritizing stability over return. This pattern mimics what we saw in our optimized portfolio, indicating that these stocks are likely the

most impactful in reducing risk. This strong concentration in GOOG across our three risk-conscious portfolios signals it as the most impactful stock for risk-adjusted performance.

Recommendations

Based on our analysis, we recommend the following actions to the Commercial Banking Corporation's department of Portfolio Pricing:

Optimized Portfolio Implementation

Implementing the optimized portfolio our team developed will allow the Commercial Banking Corporation's Portfolio Pricing department to minimize daily risk while meeting its return requirements. Our optimized portfolio achieves a minimum daily risk of 1.8% while maintaining a daily return of at least 0.18%, fulfilling both the budget and return constraints established by the Bank. By allocating approximately 69% of the portfolio to GOOG and nearly 20% to AAPL, the portfolio capitalizes on the relative stability and favorable risk-return profiles of these two stocks. The remaining allocations of 8.77% to META and 2.47% to AVGO provide modest diversification benefits, while the 0% allocation to NVDA reflects its disproportionate contribution to portfolio risk. The Bank should adopt this allocation as its primary operating portfolio, as it represents the most efficient use of capital given the return objective.

Exploring Efficient Frontier Portfolio Risk Options

In addition to the optimized portfolio, the Bank should explore the portfolio strategies identified along the efficient frontier, with particular attention to the maximum Sharpe ratio portfolio. This portfolio, with a daily risk of 2.0% and a daily return of 0.22%, represents the point on the frontier where the return earned per unit of risk is highest. It allocates 92.38% to GOOG and 7.60% to AVGO, offering a meaningful return premium over the optimized portfolio with only a modest increase in risk. We recommend that the Bank consider this portfolio when it seeks to improve return performance while still operating within a defensible risk framework. The low-risk portfolio, at 1.7% daily risk and 0.14% daily return, remains a suitable option for periods requiring risk aversion and capital preservation, diversifying across GOOG, AAPL, and META. The high-risk portfolio, concentrated entirely in AVGO at 3.4% daily risk and a daily return of 0.23%, is presented for completeness but is not recommended as a primary strategy, as it inflates risk while only slightly increasing return relative to the Sharpe ratio portfolio.

Conclusion

Our analysis has successfully delivered a robust portfolio optimization solution that meets the Commercial Banking Corporation's objectives of minimizing daily risk while achieving the required return target, and exploring efficient frontier portfolio options. Using optimization, our team identified a portfolio that minimizes daily risk to 1.8% while delivering a daily return of at least 0.18%, driven primarily by a dominant allocation to GOOG and a secondary allocation to AAPL. Additionally, the exclusion of NVDA from the optimized portfolio confirms that this stock does not contribute favorably to risk minimization under the Bank's constraints.

The efficient frontier analysis further enriches the Bank's decision-making by identifying three distinct portfolio strategies across the risk spectrum. The maximum Sharpe ratio portfolio stands out as the most compelling alternative, achieving a 0.22% daily return at 2.0% daily risk, providing a superior risk-return trade-off compared to both the low and high-risk extremes. The high-risk portfolio, while on the frontier, carries significantly more risk for a negligible increase in return. Implementation of the optimized portfolio, with the option to shift toward the maximum Sharpe ratio portfolio when risk and return objectives allow, will enable the Bank to make disciplined, data-driven investment decisions while maintaining an effective risk management strategy.